PORTFOLIO PROJECT | PYTHON · SQL · POWER BI

Customer Behavior Analysis

How can a retailer use consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

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1. Executive Summary

This project analyzes 3,900 customer purchase records from a US retailer (all 50 states, amounts in USD) to answer one business question: how can the company use shopping data to identify trends, improve customer engagement, and optimize its marketing and product strategies? The work followed three stages: the data was cleaned and explored in Python, ten business questions were answered in SQL, and the findings were presented in a four-page Power BI dashboard.

Across the dataset, total revenue is **$233,081**, the average purchase is **$59.76**, and the average review rating is **3.75 out of 5**.

Key findings

- **Revenue is steady across seasons, and Clothing leads in every one.** Fall is the strongest season ($60,018) and Summer the weakest ($55,777, about 7% lower). Clothing alone brings in 44.7% of revenue.
- **Customers are mostly male and evenly spread by age.** 68% of customers are male, and the split stays between 67% and 69% in every age group. Middle-aged customers look like the largest group only because their age band is twice as wide as the others.
- **Subscribers are a minority and do not spend more.** 27% of customers subscribe. Their average purchase ($59.49) is the same as non-subscribers' ($59.87), although they have slightly more previous purchases (26.1 vs 25.1).
- **Discounts do not lift spend or ratings.** 43% of purchases carry a discount, yet the average purchase is $59.28 with a discount and $60.13 without. Ratings are 3.74 and 3.76.
- **Discounts follow subscription.** Every one of the 1,053 subscriber purchases is discounted, against 21.9% of non-subscriber purchases. Even among non-subscribers alone, discounted purchases are not larger.
- **Shipping and payment choices are neutral.** Across six shipping types and six payment methods, average spend stays between $58 and $61 and ratings between 3.7 and 3.8.

> **Bottom line:** the data gives little support for blanket discounting and shows no spending premium for subscribers or for any shipping or payment option. The clearest commercial signals are in the product mix (Clothing and Accessories generate 76.5% of revenue) and in modest seasonal differences, so those are the most reliable places to focus.

2. Project Overview

2.1 Business question and objectives

The guiding question was: **How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?** It was split into three themes, each answered on its own dashboard page:

- **Trends:** how do purchases change from season to season, and what drives the change?

- **Customers:** who are the customers, and how engaged are they (age, gender, subscription, previous purchases)?
- **Marketing:** do discounts, shipping options, and payment choices change what customers spend and how they rate their purchase?

2.2 Approach and tools

Table 1. Project workflow

Stage	Tool	Output
Cleaning and exploration	Python (pandas)	Cleaned dataset (customer_shopping_behavior_clean.csv)
Business analysis	SQL (MySQL)	Ten queries answering specific business questions
Dashboard	Power BI	Four-page interactive dashboard (Cover, Trends, Customers, Marketing)

3. Data

3.1 Dataset overview

The cleaned dataset has **3,900 rows and 19 fields**. Each row is one purchase made by one customer, and every Customer ID appears once, so the number of customers equals the number of purchases. The data has **no date field**, so seasonality is analyzed by the season label rather than over time.

- Purchase amounts range from $20 to $100 (median $60).
- Review ratings range from 2.5 to 5.0 (mean 3.75).
- Customers are 18 to 70 years old (mean 44).
- 25 products in 4 categories (Clothing, Accessories, Footwear, Outerwear), from 50 US states.

3.2 Data dictionary

Table 2. Fields in the cleaned dataset

Field	Description	Values
customer_id	Unique customer identifier	1 to 3,900
age	Customer age	18 to 70
gender	Customer gender	Male, Female
item_purchased	Product bought	25 products
category	Product category	4 categories
purchase_amount	Purchase value in USD	$20 to $100
location	US state of the customer	50 states
size	Item size	S, M, L, XL
color	Item color	25 colors

Field	Description	Values
season	Season of purchase	Fall, Spring, Summer, Winter
review_rating	Customer rating of the purchase	2.5 to 5.0
subscription_status	Whether the customer subscribes	Yes, No
shipping_type	Shipping option chosen	6 options
discount_applied	Whether a discount was applied	Yes, No
previous_purchases	Number of earlier purchases	1 to 50
payment_method	Customer's favorite payment method	6 methods
frequency_of_purchases	How often the customer buys (text)	7 labels
age_group (derived)	Age band created during cleaning	4 groups
frequency_purchase_days (derived)	Purchase frequency in days	7 to 365

3.3 Cleaning and preparation

Cleaning was done in a Python notebook before the data was exported and loaded into MySQL for the SQL analysis.

Table 3. Data preparation steps

Step	Issue found	Action taken
Column names	Irregular names with spaces and capitals	Converted to snake_case; purchase_amount_(usd) renamed to purchase_amount
Missing values	Only review_rating had gaps: 37 nulls (0.9% of rows)	Filled with the median rating of the same category, so each product group keeps its own rating level
Age bands	Age is continuous (18 to 70)	Created age_group: young adult (18–29), adult (30–39), middle-aged (40–59), senior (60–70)
Purchase frequency	Stored as text labels	Encoded as days in frequency_purchase_days: Weekly 7, Fortnightly and Bi-Weekly 14, Monthly 30, Quarterly and Every 3 Months 90, Annually 365
Redundant field	discount_applied and promo_code_used match in every row	Dropped promo_code_used
Export	Analysis-ready dataset needed for SQL	Saved as CSV and loaded into MySQL

After cleaning, the dataset has no missing values and no duplicate customer IDs. Two modeling choices are worth stating: the seven frequency labels map to five distinct day values because Fortnightly/Bi-Weekly and Quarterly/Every 3 Months are treated as equivalent, and the four age bands are unequal in width, which matters when comparing group sizes (see Section 5.2).

4. SQL Business Analysis

Ten business questions were answered with SQL against the cleaned table. The queries use aggregation, subqueries, conditional aggregation, common table expressions (CTEs), and window functions. Results are summarized below.

Q1. Revenue by gender

What is the total revenue generated by male vs. female customers?

Table 4. Revenue by gender

Gender	Revenue	Customers	Avg purchase
Male	$157,890	2,652	$59.54
Female	$75,191	1,248	$60.25

> **Takeaway:** male customers generate 67.7% of revenue. Per purchase, women spend slightly more ($60.25 vs $59.54), so the gap is about customer volume, not basket size.

Q2. Discounted customers who still spent above average

Which customers used a discount but still spent more than the average purchase amount?

839 customers used a discount and still spent at least the average. That is exactly half (50.0%) of the 1,677 discounted purchases, which says discounts neither pull spending down nor lift it: discounted purchases are as likely to be above average as below.

Q3. Highest-rated products

Which are the top 5 products with the highest average review rating?

Table 5. Top five products by average rating

Rank	Product	Avg rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

> **Takeaway:** ratings are tightly clustered. Across all 25 products the average rating runs only from 3.62 (Shirt) to 3.86 (Gloves), so no product is a clear quality outlier.

Q4. Standard vs. Express shipping

Compare the average purchase amounts between Standard and Express shipping.

Table 6. Average purchase by shipping type

Shipping type	Customers	Avg purchase
Express	646	$60.48
Standard	654	$58.46

Express customers spend about $2 more per purchase (3.4%). The gap is small and, as Section 5.4 shows, not statistically distinguishable from chance across all six shipping options.

Q5. Do subscribers spend more?

Compare average spend and total revenue between subscribers and non-subscribers.

Table 7. Spend by subscription status

Subscription	Customers	Total revenue	Avg purchase
Subscribers	1,053	$62,645	$59.49
Non-subscribers	2,847	$170,436	$59.87

> **Takeaway:** subscribers do not spend more per purchase. Their share of revenue (26.9%) matches their share of customers (27.0%), so subscription is not currently associated with extra spend.

Q6. Products most often bought with a discount

Which 5 products have the highest percentage of purchases with discounts applied?

Table 8. Top five products by discount rate

Rank	Product	Discount rate
1	Hat	50.00%
2	Sneakers	49.66%
3	Coat	49.07%
4	Sweater	48.17%
5	Pants	47.37%

The overall discount rate is 43.0%, so these products sit 4 to 7 points above it. No product is discounted dramatically more than the rest.

Q7. Customer segments by purchase history

Segment customers into New, Returning, and Loyal based on their number of previous purchases, and show the count of each segment.

Table 9. Customer segments

Segment	Definition (previous purchases)	Customers	Share
New	1	83	2.1%
Returning	2 to 10	701	18.0%
Loyal	11 or more	3,116	79.9%

Because previous purchases run evenly from 1 to 50, almost four in five customers fall into the Loyal segment. The result is accurate for the thresholds used; a finer split of the Loyal group is suggested in Section 8.

Q8. Top three products in each category

What are the top 3 most purchased products within each category?

Table 10. Most purchased products per category (number of orders)

Category	Most purchased	Second	Third
Accessories	Jewelry (171)	Belt (161)	Sunglasses (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	n/a (only two products)

Order counts within each category are very close (Blouse and Pants tie at 171, Belt and Sunglasses at 161), so the ranking between tied products is not meaningful. Outerwear contains only two products, so it has no third place.

Q9. Are repeat buyers more likely to subscribe?

Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

Table 11. Subscription status of repeat buyers (more than 5 previous purchases)

Subscription	Repeat buyers	Share of repeat buyers
Subscribed	958	27.6%
Not subscribed	2,518	72.4%
Total	**3,476**	**100%**

> **Takeaway:** 27.6% of repeat buyers subscribe, compared with 27.0% of all customers. Repeat buying and subscribing are essentially unrelated in this data.

Q10. Revenue contribution by age group

What is the revenue contribution of each age group?

Table 12. Revenue by age group

Age group	Ages	Revenue	Share of revenue
Middle-aged	40–59	$89,741	38.5%
Young adult	18–29	$52,905	22.7%
Senior	60–70	$46,894	20.1%
Adult	30–39	$43,541	18.7%

Middle-aged customers contribute the most revenue, but this follows from the size of the band (20 years against 10 to 12 for the others). Average spend per purchase is almost identical across all four groups ($59.43 to $60.26).

5. Dashboard Findings

The Power BI dashboard has a cover page and three analytical pages. Each analytical page opens with a headline question or finding, a row of KPI cards, supporting visuals, and a short takeaway panel. This section reports what each page shows; the visuals themselves are in the dashboard.

5.1 Trends: how do purchases change from season to season?

KPI cards: revenue $233K, average purchase $59.76, average rating 3.75, 3.9K customers. A season selector filters the whole page.

Table 13. Purchases by season

Season	Purchases	Revenue	Avg purchase	Avg rating
Fall	975	$60,018	$61.56	3.73
Spring	999	$58,679	$58.74	3.79
Winter	971	$58,607	$60.36	3.75
Summer	955	$55,777	$58.41	3.73

Revenue is steady: Fall is highest ($60.0K) and Summer is lowest ($55.8K), about 7% lower. Fall leads because customers spend more per purchase ($61.56), not because there are more of them; Spring actually has the most purchases (999).

Table 14. Revenue by category and season (USD)

Category	Fall	Spring	Summer	Winter	Total	Share
Accessories	19,874	17,007	19,028	18,291	74,200	31.8%
Clothing	26,220	27,692	23,078	27,274	104,264	44.7%
Footwear	8,665	9,555	9,393	8,480	36,093	15.5%

Category	Fall	Spring	Summer	Winter	Total	Share
Outerwear	5,259	4,425	4,278	4,562	18,524	7.9%
Total	**60,018**	**58,679**	**55,777**	**58,607**	**233,081**	**100%**

- **Clothing leads revenue in every season.** Clothing in Spring ($27,692) is the strongest category–season combination.
- **Outerwear in Summer ($4,278) is the weakest combination.** Outerwear's best season is Fall ($5,259), not Winter.
- **Footwear peaks in Spring and Summer** ($9,555 and $9,393) and dips in Winter ($8,480).

Table 15. Top five products by revenue

Product	Revenue
Blouse	$10,410
Shirt	$10,332
Dress	$10,320
Pants	$10,090
Jewelry	$10,010

Four of the top five products are Clothing and one is Accessories. The gap between first and fifth place is only about 4%, so the top of the product ranking is a tight race.

5.2 Customers: who buys, and how engaged are they?

KPI cards: 3.9K customers, 27.00% subscription rate, 3.75 average rating.

Table 16. Customers by age group and gender

Age group	Ages	Customers	Share	Male share	Customers per year of age	Revenue
Young adult	18–29	878	22.5%	68.7%	73.2	$52,905
Adult	30–39	724	18.6%	67.7%	72.4	$43,541
Middle-aged	40–59	1,510	38.7%	67.4%	75.5	$89,741
Senior	60–70	788	20.2%	68.8%	71.6	$46,894

- **Mostly male.** 68% of customers (2,652 of 3,900) are male, and the share stays at 67–69% in every age group.
- **Evenly spread by age.** Ages 18 to 70 are represented about equally. Middle-aged looks largest only because its band spans 20 years; per year of age, every group holds 72 to 76 customers.
- **Subscribers are a minority.** 27% of customers subscribe (1,053 customers), and they have slightly more previous purchases than non-subscribers in every age group.

Table 17. Average previous purchases by age group and subscription

Age group	Non-subscribers	Subscribers
Young adult	24.2	26.0
Adult	24.6	25.5
Middle-aged	25.4	26.0
Senior	25.8	26.9

Table 18. Favorite payment methods

Payment method	Share of customers	Avg purchase
PayPal	17.4%	$59.25
Credit Card	17.2%	$60.07
Cash	17.2%	$59.70
Debit Card	16.3%	$60.92
Venmo	16.3%	$58.95
Bank Transfer	15.7%	$59.71

Payments do not differentiate customers. All six methods fall between 15.7% and 17.4% of customers, with average spend within about $2 of each other.

5.3 Marketing: do discounts, shipping, and payment change spend and ratings?

KPI cards: 43.00% discount usage, $59.28 average purchase with a discount, $60.13 without.

Table 19. Discounted vs. non-discounted purchases

Group	Purchases	Share	Avg purchase	Avg rating
Discounted	1,677	43.0%	$59.28	3.74
Not discounted	2,223	57.0%	$60.13	3.76

Table 20. Discount use by subscription status

Group	Purchases	With discount	Without discount	Discount rate
Subscribers	1,053	1,053	0	100.0%
Non-subscribers	2,847	624	2,223	21.9%

- **Discounts do not lift spend or ratings.** Discounted purchases average $59.28 against $60.13 without a discount, and ratings are 3.74 against 3.76.
- **Discounts follow subscription.** All subscriber purchases carry a discount, against 21.9% for non-subscribers, so the two groups are not a clean comparison. Looking at non-subscribers alone removes that overlap, and the result is the same: $58.92 with a discount against $60.13 without.

Table 21. Shipping type: customers, spend, and rating

Shipping type	Customers	Avg purchase	Avg rating
2-Day Shipping	627	$60.73	3.77
Express	646	$60.48	3.77
Next Day Air	648	$58.63	3.72
Store Pickup	650	$59.89	3.71
Standard	654	$58.46	3.82
Free Shipping	675	$60.41	3.72
Total	**3,900**	**$59.76**	**3.75**

Shipping and payment are neutral. Across six shipping types and six payment methods, average spend stays between $58 and $61 and ratings between 3.7 and 3.8. Standard shipping has the highest rating (3.82) and the lowest average spend ($58.46), but the differences are small.

6. Recommendations

These recommendations follow from the findings above. Because the data is a single snapshot, they are best treated as hypotheses to test rather than proven levers.

1. **Re-examine blanket discounting.** Discounts apply to 43% of purchases without raising spend or ratings. Before continuing broad discounts, test targeted promotions against a control group to see whether any segment responds.
2. **Review what subscription offers.** Subscribers receive a discount on every purchase yet spend the same as non-subscribers and are only slightly more experienced buyers. Benefits aimed at retention and purchase frequency may be a better fit than discounts.
3. **Keep Clothing and Accessories at the center of the assortment.** Together they produce 76.5% of revenue and lead in every season. Merchandising and inventory planning should protect these two categories year-round.
4. **Plan around the seasonal rhythm.** Fall is the strongest season, driven by higher spend per purchase, and Summer is the softest. Summer is the natural window for a campaign, while Footwear is well placed for Spring and Summer pushes.
5. **Grow the female customer base if possible.** Women are only 32% of customers but spend as much per purchase as men ($60.25 vs $59.54), so each additional female customer is worth as much as a male one.
6. **Treat shipping and payment as operational decisions.** Neither shows a spending or rating advantage, so keep a broad mix of options and prioritize cost and reliability rather than expecting a sales lift.

7. Limitations

- **No date field.** Seasonality is based on the season label, so growth, trends over time, and campaign timing cannot be studied.
- **One purchase per customer.** Repeat behavior and customer lifetime value cannot be measured directly. Previous purchases and purchase frequency are summary fields, not transaction history.
- **Correlation, not causation.** The data cannot show whether a discount or a subscription caused a change in spending.
- **Overlapping variables.** Discounts and subscription are linked (all subscriber purchases are discounted), which limits how cleanly each can be evaluated.
- **Little variation to explain.** Most variables are close to evenly distributed (purchase amounts span $20 to $100 with a median of $60; previous purchases span 1 to 50), so no single factor explains much of the difference in spending. The conclusions should be validated on real transaction data before driving decisions.
- **Imputation.** 37 missing ratings (0.9%) were filled with category medians. This is a small share and unlikely to change any result.

8. Next Steps

- **Add a predictive model.** A simple model (for example logistic regression or gradient boosting) predicting subscription or above-average spend would test whether combinations of features carry signal that single-variable comparisons miss.
- **Add a geographic view.** State-level revenue is not yet in the dashboard (highest: Montana at $5,784; lowest: Kansas at $3,437), and should be read alongside customer counts per state.
- **Explore product attributes.** Size and color were not analyzed and could inform assortment decisions.
- **Collect dates and transaction history** to allow time-series analysis, cohort analysis, and true customer lifetime value.